Type MS · Explain how mosquitoes and houseflies spread disease

Answer 1 of 5 · Question 8 part a · 3 marks

2019 JUN · Paper 01 · extract 1 of 2 · find part a in this mark scheme extract

8 (a)	Three from: • (parasite) carried by mosquito/mosquito is a vector; • mosquito bites human; • parasite/infected blood drawn/sucked up into mosquito; • infected blood transmitted to other people;		Max 3
(b)(i)	• fewer deaths caused by *P.vivax* in R than Q • (but) more deaths caused by *P.vivax* in R than in Q as a proportion of the total; • 190 000 fewer deaths caused by *P.vivax* in R/300 less deaths in Q caused by *P.vivax*; • 0.5% deaths caused by *P.vivax* in Q/62.5% of deaths caused by *P.vivax* in R;		1 1 1
(ii)	more mosquitoes in one region than another/climate favours breeding of mosquitoes/more dense population of people/better health care/use of (named) preventative measures;		1
(c)	Three from: Fewer people with malaria/reduced incidence of malaria; Immunity/antibodies against parasite/herd immunity; Reduction in population of/death of parasite; Less transmission (from one person to another);		Max 3
(d)	Three from: (sexual reproduction) • gives rise to variation in offspring; • variation provides a survival advantage; • parasite more likely to survive in a changing environment; (asexual reproduction) • parasite can reproduce faster; • no need to find a mate/only one parent needed; • larger number of offspring produced;	Answer must contain at least one advantage of each method	Max 3

Type MS · Explain how mosquitoes and houseflies spread disease

Answer 1 of 5 · Question 8 part a · 3 marks

2019 JUN · Paper 01 · extract 2 of 2 · find part a in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Explain how mosquitoes and houseflies spread disease

Answer 2 of 5 · Question 4 part b.i · 1 mark

2023 JAN · Paper 01 · extract 1 of 2 · find part b.i in this mark scheme extract

Number					
4	(a)	(i)	cell wall/capsule/slime coat;		1
		(ii)	55 ÷ 30 000 = 0.0018: 1.8/1.83/1.833;		2
	(b)	(i)	more densely/higher populated/more people/overcrowded/more waste/rubbish/sewage;		1
		(ii)	• fly lands on faeces (of person with a stomach ulcer); • transfers the bacteria to food; • person ingests the contaminated food;		3
	(c)	(i)	Two from: • high/medium levels in less well developed/poorer countries; • where sewage treatment is less effective/water not clean/poor sanitation; • more densely/highly populated; • healthcare/vaccines to treat infected people is less effective/available;	accept reverse arguments	2
		(ii)	Five from: • bacteria mutate/acquire resistant plasmids; • people do not finish the course/over-use of antibiotics; • resistant bacteria survive (and non-resistant bacteria die); • (resistant bacteria) reproduce/multiply; • offspring are antibiotic resistant/not affected by antibiotics; • antibiotic that will be effective is needed/used; • more serious infection occurs;		5

Total 14 marks

Type MS · Explain how mosquitoes and houseflies spread disease

Answer 2 of 5 · Question 4 part b.i · 1 mark

2023 JAN · Paper 01 · extract 2 of 2 · find part b.i in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Explain how mosquitoes and houseflies spread disease

Answer 3 of 5 · Question 4 part b.ii · 3 marks

2023 JAN · Paper 01 · extract 1 of 2 · find part b.ii in this mark scheme extract

Number					
4	(a)	(i)	cell wall/capsule/slime coat;		1
		(ii)	55 ÷ 30 000 = 0.0018: 1.8/1.83/1.833;		2
	(b)	(i)	more densely/higher populated/more people/overcrowded/more waste/rubbish/sewage;		1
		(ii)	• fly lands on faeces (of person with a stomach ulcer); • transfers the bacteria to food; • person ingests the contaminated food;		3
	(c)	(i)	Two from: • high/medium levels in less well developed/poorer countries; • where sewage treatment is less effective/water not clean/poor sanitation; • more densely/highly populated; • healthcare/vaccines to treat infected people is less effective/available;	accept reverse arguments	2
		(ii)	Five from: • bacteria mutate/acquire resistant plasmids; • people do not finish the course/over-use of antibiotics; • resistant bacteria survive (and non-resistant bacteria die); • (resistant bacteria) reproduce/multiply; • offspring are antibiotic resistant/not affected by antibiotics; • antibiotic that will be effective is needed/used; • more serious infection occurs;		5

Total 14 marks

Type MS · Explain how mosquitoes and houseflies spread disease

Answer 3 of 5 · Question 4 part b.ii · 3 marks

2023 JAN · Paper 01 · extract 2 of 2 · find part b.ii in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Explain how mosquitoes and houseflies spread disease

Answer 4 of 5 · Question 8 part b.ii · 1 mark

2019 JUN · Paper 01 · extract 1 of 2 · find part b.ii in this mark scheme extract

8 (a)	Three from: • (parasite) carried by mosquito/mosquito is a vector; • mosquito bites human; • parasite/infected blood drawn/sucked up into mosquito; • infected blood transmitted to other people;		Max 3
(b)(i)	• fewer deaths caused by *P.vivax* in R than Q • (but) more deaths caused by *P.vivax* in R than in Q as a proportion of the total; • 190 000 fewer deaths caused by *P.vivax* in R/300 less deaths in Q caused by *P.vivax*; • 0.5% deaths caused by *P.vivax* in Q/62.5% of deaths caused by *P.vivax* in R;		1 1 1
(ii)	more mosquitoes in one region than another/climate favours breeding of mosquitoes/more dense population of people/better health care/use of (named) preventative measures;		1
(c)	Three from: Fewer people with malaria/reduced incidence of malaria; Immunity/antibodies against parasite/herd immunity; Reduction in population of/death of parasite; Less transmission (from one person to another);		Max 3
(d)	Three from: (sexual reproduction) • gives rise to variation in offspring; • variation provides a survival advantage; • parasite more likely to survive in a changing environment; (asexual reproduction) • parasite can reproduce faster; • no need to find a mate/only one parent needed; • larger number of offspring produced;	Answer must contain at least one advantage of each method	Max 3

Type MS · Explain how mosquitoes and houseflies spread disease

Answer 4 of 5 · Question 8 part b.ii · 1 mark

2019 JUN · Paper 01 · extract 2 of 2 · find part b.ii in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Explain how mosquitoes and houseflies spread disease

Answer 5 of 5 · Question 4 part c.i · 2 marks

2023 JAN · Paper 01 · extract 1 of 2 · find part c.i in this mark scheme extract

Number					
4	(a)	(i)	cell wall/capsule/slime coat;		1
		(ii)	55 ÷ 30 000 = 0.0018: 1.8/1.83/1.833;		2
	(b)	(i)	more densely/higher populated/more people/overcrowded/more waste/rubbish/sewage;		1
		(ii)	• fly lands on faeces (of person with a stomach ulcer); • transfers the bacteria to food; • person ingests the contaminated food;		3
	(c)	(i)	Two from: • high/medium levels in less well developed/poorer countries; • where sewage treatment is less effective/water not clean/poor sanitation; • more densely/highly populated; • healthcare/vaccines to treat infected people is less effective/available;	accept reverse arguments	2
		(ii)	Five from: • bacteria mutate/acquire resistant plasmids; • people do not finish the course/over-use of antibiotics; • resistant bacteria survive (and non-resistant bacteria die); • (resistant bacteria) reproduce/multiply; • offspring are antibiotic resistant/not affected by antibiotics; • antibiotic that will be effective is needed/used; • more serious infection occurs;		5

Total 14 marks

Type MS · Explain how mosquitoes and houseflies spread disease

Answer 5 of 5 · Question 4 part c.i · 2 marks

2023 JAN · Paper 01 · extract 2 of 2 · find part c.i in this mark scheme extract

Question number	Answer	Notes	Marks